

Education

University of Central Florida

August 2022 - December 2026

Bachelor of Engineering in Computer Engineering; GPA: 3.3

Orlando, FL

- Relevant Coursework: Engineering Computation and Analysis, Data Structures and Algorithms, Digital Systems, Operating Systems, Machine Learning, Embedded Systems, Computer Architecture, Electronics.

Experience

VR Orlando Rentals, Orlando FL

Jan 2023 - May 2023

Coding Teacher

- Delivered programming/engineering lessons to K-12 students using Scratch and VR
- Adapted instructions for 10-30 students at varying skill levels
- Collaborated with program leadership to coordinate classroom logistics and ensure day-to-day operations across multiple classrooms.

JLB Services, Manchester, United Kingdom

March 2025 - Present

Freelance development

- Designed and built a custom delivery-service booking web app using React.js and RESTful APIs
- Implemented backend logic and database integration (via Supabase) to store user orders, manage user sessions, and enable retrieval of past bookings

Projects

KnightHacks Competition: Project Artemis

January 2025 - April 2025

Team Leader/Embedded Systems Developer

- Implemented a UART link from Arduino to Raspberry Pi for real-time turret control
- Calibrated baud rate (115200 bps) and buffering to meet update-rate targets while preventing overruns; added simple framing + checksum to avoid ghost inputs
- Collaborated in a cross-functional team to align hardware-software communication and optimize responsiveness in object detection and servo actuation.

Language Learning App

January 2025 - Present

Full-stack developer

- Developed frontend with HTML, CSS, Javascript and backend with Node.js/MySQL
- Integrated AI-based NLP to extract semantic meaning from Spanish and Russian text interactive features.
- Added account creation and personalized word-list storage features for user progress tracking

Simplified TCP Protocol Simulator

April 2025 - May 2025

Computer Communications and Network

- Implemented a reliable transport protocol with sequence numbers, ACKs, retransmission timers, and out-of-order packet buffering over a simulated unreliable channel
- Built sender and receiver modules supporting dynamic window sizing and congestion handling, demonstrating TCP/IP reliability mechanisms

FPGA Design and RISC-V CPU Integration

March 2024 - August 2025

Digital Systems, Basys3, Vivado, LiteX

- Designed and implemented digital circuits on FPGA, including 8-bit adders, multiplexers, and RAM/ROM modules
- Gained hands-on experience with Verilog, synthesis, and hardware verification workflows
- Integrated and deployed a RISC-V soft CPU onto a BASYS3 board with SoC addition

Sharedle Wordle Clone

August 2025 - November 2025

Processes of Object Oriented Software Design - Backend Developer

- Developed a full-stack multi-platform Wordle-style word guessing game with backend APIs for word validation, score tracking, and user authentication
- Designed RESTful endpoints and integrated frontend with backend via JSON-based communication.

Skills

Skills: Verilog, C, C++, Python, Java, MATLAB, MIPS Assembly, HTML/CSS/JS, React, Supabase, Docker, Git/GitHub, Linux, Vivado, LTSpice

Languages: English, Spanish, Russian

Hardware and Lab Tools: Oscilloscopes, Multimeters, Spectrum analyzers, Function Generators